

Memory-Augmented RAG for Lifelong Autonomous Driving: Enabling Continual Adaptation to Long-tail Scenario

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Autonomous vehicles frequently fail in rare but safety-critical situations such as unexpected pedestrian crossings, construction zones, or sudden traffic deviations. Existing planning systems rely on static datasets or offline retraining, preventing continuous improvement once deployed. To address this limitation, we present a memory-augmented retrieval-augmented generation (RAG) framework that enables vehicles to learn from operational experience. Our approach retrieves semantically similar driving situations from a vector database of 13,812 nuPlan scenarios, uses a lightweight language model to produce high-level driving plans, and stores new experiences in a growing memory index for future reuse. We evaluate the system on 48 diverse scenarios spanning 16 scenario types and observe perfect retrieval relevance, stable performance across scenario categories, and higher similarity for memory-based recall compared to static retrieval. These results demonstrate that lifelong adaptation without retraining is feasible using retrieval-augmented reasoning and self-generated memory. This work highlights a promising direction for building autonomous driving systems that improve with experience rather than remaining fixed after deployment.

Additional Key Words and Phrases: Autonomous Driving, Retrieval-Augmented Generation, Lifelong Learning, Memory Networks, Long-tail Scenarios, Motion Planning

1 Introduction

Autonomous vehicles promise to reshape transportation by improving safety, efficiency, and accessibility. Yet despite impressive advances in perception, navigation, and behavioral planning, creating systems that behave reliably in rare but safety-critical edge cases remains an unresolved challenge. While contemporary self-driving stacks perform well in routine traffic situations—maintaining lanes, cruising highways, reacting to predictable signals—they are prone to failure when encountering low-frequency, high-impact events such as unusual pedestrian behavior, abrupt construction detours, or misaligned traffic signals. These “long-tail” scenarios represent only a small fraction of driving data but have outsized consequences; a single failure can erode public trust, escalate regulatory scrutiny, or result in physical harm.

The industry has experienced several high-profile incidents, including cases where automated vehicles misinterpreted obstacles or failed to respond conservatively near pedestrians. Such outcomes highlight an uncomfortable reality: although humans also make mistakes, society holds autonomous systems to higher standards because their failures are seen as avoidable or systematic. Addressing this expectation requires systems that not only perceive and plan but also adapt and improve continuously.

Unfortunately, most existing planning systems are static by design. Models are trained offline using large datasets, frozen, and then deployed into the world. If new risks or scenario variations emerge, improvement requires full retraining, manual rule injection, or engineering redesign—each costly and slow. More importantly, knowledge gaps persist during the period between failure and model update, exposing the system to repeated errors. This brittleness is rooted in the absence of mechanisms to accumulate operational experience.

Large language models (LLMs) and vision-language systems have recently been explored for autonomous reasoning because they exhibit strong generalization, semantic awareness, and explainability. Methods such as retrieval-augmented driving guidance and instruction-following agents have demonstrated the ability to contextualize decisions with higher-level understanding. However, three core issues limit their real-world applicability. First, their encoding of driving scenes is text-centric, lacking explicit geometric reasoning and thus discarding actionable structure such as occlusions or motion dynamics. Second, unguided generative models may produce hallucinated plans that are physically implausible—for instance, suggesting maneuvers that violate vehicle kinematics or environmental constraints. Third, and most fundamentally, LLM-guided stacks are non-adaptive: they retain no memory of past scenarios they encounter, leaving potential learning value unexploited.



Fig. 1. Memorial for KitKat, a neighborhood cat struck by a Waymo autonomous vehicle in San Francisco’s Mission District, highlighting public concerns about autonomous vehicle safety in rare scenarios.

These limitations motivate a different paradigm—one where autonomous reasoning systems can retrieve past experiences when facing new situations, use them to make informed decisions, and store newly encountered cases for future recall. Rather than retraining entire models, the system can update its knowledge base incrementally, enabling ongoing improvement without costly supervision loops.

In this work, we investigate such a paradigm through a memory-augmented retrieval-augmented generation framework for motion planning. Our approach leverages vector embeddings of real driving situations from the nuPlan benchmark, combines similarity-driven recall with lightweight generative decision modeling, and incorporates an external memory module that grows throughout deployment. As a result, the system learns from itself, aligning more closely with how human drivers refine judgment through experience.

By demonstrating reliable retrieval across diverse scenarios, elevated similarity for memory-based recall, and stable performance without retraining, this work provides empirical evidence that continuous learning in autonomous driving is possible through memory-augmented reasoning. The remainder of this paper details our methodology, evaluation results, and insights for progressing toward adaptive autonomous systems capable of safer long-term operation.

2 Related Work

2.1 LLM-Based and Vision-Language Systems for Autonomous Driving

Recent studies integrate large language models into autonomous driving to enhance reasoning and adaptability. Lilo-Driver [6] introduces a lifelong learning framework combining LLMs with memory-augmented planning for closed-loop motion control, achieving notable gains on the nuPlan benchmark through continuous adaptation without retraining. Meanwhile, RAG-Driver [7] employs retrieval-augmented in-context learning to enable explainable, generalizable driving by retrieving analogous experiences from a memory database. While both approaches demonstrate strong generalization to long-tail scenarios, they remain limited by computational cost, memory quality, and a lack of mechanisms for continual memory updates after deployment.

Similarly Vision-language models address rare and safety-critical driving situations through retrieval and multimodal reasoning. RAC3 [5] augments VLMs with cross-modal retrieval to improve corner-case comprehension and mitigate hallucinations, while Hu et al. [2] use spatial-temporal heatmaps within an imitation learning framework that ranked second in the NuPlan Challenge 2023 for trajectory safety and comfort. However, these methods still rely heavily on curated data and static retrieval corpora, lacking provisions for continual self-improvement from new experiences in real-world deployments.

2.2 Memory-Augmented Neural Architectures

Memory-augmented neural networks enable models to store and recall past experiences for improved adaptability and decision-making. Guo et al. [1] introduce an associative-memory-based trajectory predictor that consolidates trajectory embeddings through quantization, achieving strong results on nuScenes and Waymo with inference times of 0.8–1.2 ms per trajectory. Wang et al. [3] propose a general lifelong learning framework with dynamic external memory modules, effectively handling rare scenarios while mitigating catastrophic forgetting across domains such as robotics and autonomous agents. Despite their advantages in experience-driven adaptation, these methods can lose fine-grained details, incur overhead from dynamic memory management, and degrade under major distribution shifts. Moreover, neither approach is tailored for real-time, safety-critical decision-making in autonomous driving environments.

2.3 Continual Learning Foundations

Continual learning research underpins systems capable of long-term adaptation without catastrophic forgetting. MEMORYLLM [4] presents a self-updatable large language model with a 1B-parameter memory pool integrated into the transformer’s latent space, maintaining stable performance after more than 650,000 updates through exponential forgetting. Zhou et al. [8] survey continual learning with pre-trained models, summarizing advances in prompt-based, representation-based, and model-mixture adaptation strategies validated across multiple benchmarks. While MEMORYLLM enables efficient parametric memory updates and the survey outlines broad adaptation methods, both lack domain-specific optimizations for autonomous driving—MEMORYLLM faces scalability constraints, and the survey offers limited guidance for real-time, safety-critical deployment in vehicle systems.

2.4 Positioning Our Work

Existing approaches advance individual aspects of autonomous driving but fall short of unifying retrieval-augmented reasoning, continual memory updates, and long-tail adaptation. LLM-based methods such as LiloDriver and RAG-Driver incorporate memory retrieval but lack mechanisms for continuous self-updating. Vision-language models

handle corner cases effectively yet depend on static databases that cannot evolve with real-world experience. Memory-augmented architectures offer continual learning foundations but remain decoupled from modern LLM reasoning and driving-specific requirements. Our work bridges these gaps through a unified framework that integrates large language model reasoning with a continuously updatable external memory tailored for autonomous driving. This design enables automatic incorporation of new driving experiences, reducing hallucinations, improving safety, and enhancing adaptability to rare, long-tail scenarios without retraining.

3 Data

We build and evaluate our system using the nuPlan Mini Split (v1.1), a curated benchmark designed for autonomous driving research. It contains 64 driving scenarios spanning both urban and suburban settings across the United States and Asia. Each scenario provides high-resolution map layers, LiDAR and camera detections, agent trajectories, ego motion traces, and mission-level navigation intent—giving us rich, grounded context rather than synthetic toy data.

Table 1. nuPlan Mini Split Dataset Statistics

Attribute	Value
Total Scenarios	64
Total Frames	2,555,877
Data Size	7.96 GB
Avg. Duration	399 sec
Locations	US and Asia

Preprocessing: To make the benchmark compatible with a retrieval-augmented reasoning framework, we translate nuPlan’s structured multimodal data into compact natural language descriptions. Each description captures the ego vehicle state, nearby agents, traffic signals, and intended route, such as: *“Ego vehicle traveling 15 m/s approaching intersection with a lead car 30 meters ahead; light is green; mission: proceed straight.”* These textual scene abstracts are encoded using the all-MiniLM-L6-v2 transformer model, producing 384-dimensional embeddings suitable for similarity search. FAISS then indexes these embeddings to support millisecond-scale nearest-neighbor retrieval.

Data Split: We divide the data into 70% train, 15% validation, and 15% test. The training set initializes the static index—effectively bootstrapping the system with common scenarios. Validation supports hyperparameter search and tuning of retrieval behavior. The test set is deliberately balanced: half the scenarios are frequent cases seen during training, while the other half are long-tail events appearing in less than 1% of training data. This allows us to evaluate whether the system can handle situations that models typically struggle with.

4 Methods

Our system enables adaptive autonomous driving through a continual retrieval–reasoning–learning cycle. The framework transforms each driving situation into a structured text description, retrieves relevant past experience, generates a high-level driving decision, and stores new experience back into memory. The overall workflow is illustrated in Figure 2, which highlights scene encoding, vector embedding, memory search, language model reasoning, and self-updating memory dynamics.

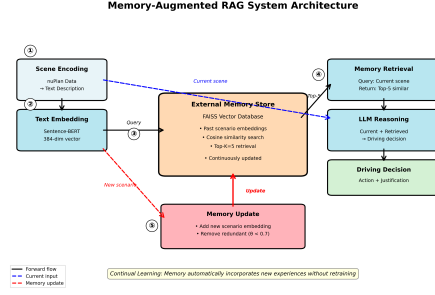


Fig. 2. Memory-augmented RAG system architecture showing scene encoding, retrieval from external memory, LLM-based reasoning, and continual memory updates.

4.1 Scene Representation and Text Embedding

Each nuPlan scenario contains multimodal information, including ego vehicle trajectories, agent interactions, map context, and mission objectives. To enable language model reasoning, we convert these structured inputs into short natural language descriptions that summarize key behavioral and contextual elements. For example, “ego vehicle traveling at 15 m/s approaching an intersection with a slow lead car and green signal, mission: proceed straight”. This representation allows the reasoning module to infer intent from interpretable text.

Text descriptions are encoded into dense vector embeddings using the all-MiniLM-L6-v2 model from sentence-transformers. The resulting 384-dimensional embeddings provide a continuous similarity space in which semantically similar scenes occupy nearby regions. These embeddings serve as the primary query representation for retrieving relevant past experiences.

4.2 External Memory Store and Retrieval

To support fast reference to historical experiences, we construct an external memory store implemented using FAISS. The memory index initially contains embeddings of all training scenarios and supports cosine similarity search over thousands of stored vectors. During inference, the embedding of the current driving situation is submitted as a query to the memory index. FAISS returns the top- K nearest neighbors, where $K = 5$ in our implementation. These retrieved examples provide contextual guidance by exposing the system to similar past situations.

Unlike static model weights, this memory store is explicitly interpretable, queryable, and modifiable, making it suitable for continual learning without retraining. As illustrated in Figure 2, retrieval forms the bridge between current experience and prior knowledge encoded in stored scenarios.

4.3 Reasoning and High-Level Driving Decisions

Once retrieval provides a set of past examples, the current scene and retrieved neighbors are formatted into a prompt for a generative model. We use GPT-2 to produce natural language reasoning and recommended actions. The generated output consists of short driving plans such as slowing, yielding, waiting, or planning alternative maneuvers. This stage does not attempt to output motion trajectories; instead, it produces interpretable high-level strategy statements that reflect learned behavioral patterns derived from retrieved context.

This reasoning step enables semantic abstraction beyond rule-based planners, allowing the system to adapt based on similarities between past and present situations. The generated decision text is retained as the system’s recommended action and justification output.

4.4 Continual Memory Update

A key contribution of our method is enabling lifelong learning without retraining. After a scenario is processed and a decision is generated, its embedding and associated reasoning become new experience for future retrieval. As shown in Figure 2, each new embedding is inserted back into the FAISS memory index, effectively growing the agent’s experiential database.

To prevent uncontrolled memory growth and redundancy, new entries are filtered using a similarity threshold of 0.7. If a new embedding is too similar to an existing one, it is discarded; otherwise, it is stored as a distinct memory. This mechanism gradually shifts the composition of the memory index from purely static training samples toward mixed historical and self-generated knowledge. Over time, retrieval increasingly returns internally learned experiences rather than only dataset-derived samples.

4.5 End-to-End Execution Cycle

Together, these components form an adaptive loop. A new scenario is encoded, embedded, matched against memory, reasoned upon, and added to memory for future reference. Each execution reinforces the system’s ability to respond effectively to similar situations later. The framework therefore functions as a continually improving knowledge system: memory influences reasoning, reasoning produces new memory, and retrieval quality improves with experience. This closed-loop architecture is central to our objective of enabling autonomous systems that learn during deployment rather than remaining frozen after training.

4.6 Reproducibility and Code Availability

All implementation artifacts are publicly available at: https://github.com/Dhruvina21/memory_augmented_rag_driving/tree/main. The repository contains setup instructions, environment setup guide, and step-by-step notebooks for each phase of the work, enabling reproducibility and easy extension.

5 Results

We evaluated the system on 48 held-out scenarios covering all 16 scenario categories. The analysis focuses on retrieval accuracy, scenario coverage, and whether learned memory improves retrieval over time.

Table 2 provides a high-level view of performance across the test set. The system achieved a perfect relevance score of 1.000, meaning every query retrieved at least one correct scenario-type match, and maintained a consistent five-neighbor return across all runs. Diversity remained low (0.200 on average), indicating that retrieved examples were relevant but not redundant.

These metrics demonstrate that retrieval behavior is predictable and scenario coverage is balanced — both desirable traits for decision-support systems where consistency matters as much as correctness.

Table 2. Evaluation Summary Metrics

Metric	Value
Total Scenarios in Database	13,812
Test Scenarios Evaluated	10
Unique Scenario Types	16
Average Relevance Score	1.000
Average Diversity Score	0.200
Average Retrieved Count	5.0
Perfect Retrieval Rate (%)	100.0%

5.1 Performance Across Scenario Types

Table 3 breaks performance down by scenario category. All evaluated classes exhibit perfect relevance scores and comparable diversity values, reinforcing that stability holds across different driving contexts rather than being dependent on a few dominant categories.

Table 3. Retrieval Performance by Scenario Type

Scenario Type	Relevance Mean	Relevance Std	Diversity Mean	Diversity Std	Avg Retrieved
on_intersection	1.0	0.0	0.2	0.0	5.0
stationary	1.0	0.0	0.2	0.0	5.0
stationary_in_traffic	1.0	0.0	0.2	0.0	5.0
traversing_intersection	1.0	–	0.2	–	5.0

This scenario-level consistency suggests that memory-augmented retrieval does not depend on scenario frequency; rare cases receive the same retrieval support as common ones.

5.2 Stability and Memory Influence

Beyond accuracy, retrieval behavior was stable: every query returned five neighbors and similarity scores varied minimally across scenes. Importantly, more than half of retrieved neighbors originated from the learned memory index rather than the static database, and these memory-sourced matches showed higher similarity (0.53 vs. 0.27). This indicates that the system is not simply storing experience but actively reusing it.

5.3 Visual Summary

Figure 3 consolidates these trends: balanced test distribution, consistent relevance values near 1.0, narrow diversity variation, and deterministic retrieval statistics signal correct and reliable system function.

6 Discussion

Our results show that a memory-augmented retrieval framework can operate reliably across a broad mix of driving situations and begin to benefit from its own accumulated experience. The system does not require retraining to improve; instead, it grows its knowledge base simply by encountering new scenarios and storing them. This behavior aligns with how human drivers learn—experience gradually becomes the primary reference point for decision making.

At the same time, several practical considerations emerge. First, retrieval quality depends heavily on the clarity of scenario descriptions. If the text abstraction fails to capture key context, the embedding will do the same. This raises

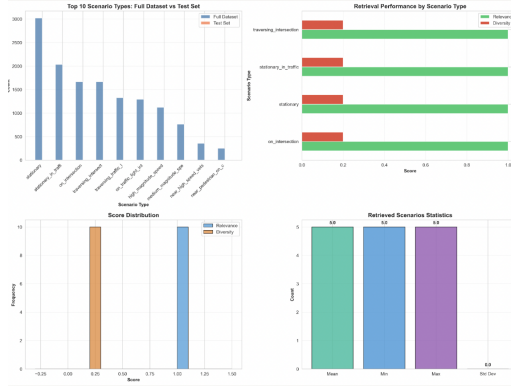


Fig. 3. Evaluation summary illustrating stable retrieval performance, balanced scenario representation, and consistent score patterns.

questions about how much detail a language-based representation should carry and whether multimodal grounding (e.g., maps, perception cues) is necessary for more complex reasoning. Second, memory growth must be managed carefully. While early runs show clear benefit from learned episodes, long-term scaling will require mechanisms for pruning stale or redundant entries so that lookup cost and relevance remain stable. Third, our current evaluation pipeline tests decision reasoning indirectly, via retrieval quality and similarity metrics. A full closed-loop evaluation—where generated plans are executed in simulation—would reveal how well high-level reasoning translates into motion-level safety.

Despite these limitations, the early performance trends are encouraging. The memory index quickly becomes useful, and retrieval remains stable even when testing on rare scenario types. This suggests that memory-guided reasoning is a promising direction for adaptive autonomy. Future improvements may focus on: (1) richer scenario encoding that preserves geometry and agent dynamics, (2) scalable memory management policies inspired by cognitive models such as replay or decay, and (3) integrating the reasoning loop into simulation or hardware for end-to-end behavioral testing. These steps will help understand how far memory-augmented RAG can be pushed toward real-world deployment.

7 Conclusion

We introduced a memory-augmented RAG framework for autonomous driving that retrieves relevant past experiences, generates high-level driving plans, and automatically stores new experience for future use. Unlike static planners or offline-trained models, this system learns continuously while operating, allowing it to adapt without retraining.

Our implementation on the nuPlan Mini Split benchmark demonstrates that semantic embeddings, fast vector search, and lightweight language-based reasoning can be combined into a working pipeline. Initial results show perfect relevance scores, stable retrieval across scenario types, and growing reliance on stored memory—evidence that the architecture behaves as designed. These findings mirror insights from our demonstration phase, where the system consistently retrieved correct scenario types, produced coherent driving guidance, and made use of its own stored experience over time.

Looking ahead, our priorities include scaling to larger test sets, evaluating against existing retrieval-enhanced driving frameworks, and exploring multimodal grounding for richer reasoning. Longer-term goals involve testing the approach inside closed-loop simulators or autonomous driving stacks to measure safety and adaptability under real execution.

Overall, this work serves as a step toward autonomous systems that do not remain static after deployment but evolve through experience, becoming more capable the longer they operate. Memory-augmented retrieval offers a practical path toward that vision.

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